

Multiple Choice (10 marks)

1. Let A be the set of all ordered pairs of integers (m, n) such that $7m + 12n = 22$. What is the greatest negative number in the set $B = \{m + n : (m, n) \in A\}$?
 - (a) -5
 - (b) 0
 - (c) -3
 - (d) -7
 - (e) -4
2. A model of a park was built on a scale of 1.5 centimeters to 50 meters. If the distance between two trees in the park is 150 meters, what is this distance on the model?
 - (a) 6 centimeters
 - (b) 150 centimeters
 - (c) 0.5 centimeter
 - (d) 9 centimeters
 - (e) 225 centimeters
3. Let V and W be 4-dimensional subspaces of a 7-dimensional vector space X . Which of the following CANNOT be the dimension of the subspace $V \cap W$?
 - (a) 0
 - (b) 1
 - (c) 9
 - (d) 8
 - (e) 2
4. Solve $13 \text{ over } 4 = x \text{ over } 7$.
 - (a) 22.25
 - (b) 23.25
 - (c) 25.25
 - (d) 23.75
 - (e) 21.75
5. If all the values of a data set are the same, all of the following must equal zero except for which one?

- (a) Mean
- (b) Standard deviation
- (c) Variance
- (d) Interquartile range
- (e) Quartile deviation

True / False (10 marks)

Answer True or False

6. True or False: The sum of the second partial derivatives of u with respect to x and y equals 2.0.
7. True or False: The arc length of $y = (1/4)x^4$ over the interval $[1,2]$ calculated using the Trapezoidal Rule T_5 is 4.322.
8. True or False: The fraction $\frac{4}{\sqrt{108}+2\sqrt{12}+2\sqrt{27}}$ simplifies to $\frac{1}{2\sqrt{3}}$.
9. True or False: Every group of order 159 is cyclic, as it is the product of two distinct primes and thus must have a non-cyclic structure.
10. True or False: The limit of $3n(a_n)^3$ as n approaches infinity is 1.0.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the value of the integral $\int_{|z|=r} \frac{f(z)}{(z-z_1)(z-z_2)} dz$ for a bounded entire function f and points z_1, z_2 within the ball $B(0, r)$. Justify your answer.
12. Determine the value of N that satisfies the equation $\frac{4^{(3^3)}}{(4^3)^3} = 2^N$. Provide a detailed analysis of your solution process.

End of Paper